

REAL-TIME SCORE FOLLOWING IN SHEET MUSIC IMAGES AS SEQUENTIAL CANDIDATE SELECTION

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ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. The two sides are different kinds of object, the performance audio and the score an image. The established response is to convert one into the other and align within a single modality, either by turning the page into symbols with optical music recognition or by turning the audio into notes with transcription, and each detour inherits the errors of its conversion. Sheet-image score following converts neither side and learns the correspondence directly, and its strongest systems condition a visual detector on the audio. What limits them is annotation, since performances aligned to notehead coordinates are costly enough that nearly all training data is synthesized, so these systems work on synthesized audio and lose much of their accuracy on a real instrument in a real room. Closing that gap is the field’s central open problem, and every remedy proposed for it has treated the gap as a failure of perception. We present evidence that it is not, since on real recordings the detector already proposes a correct position at every scored onset and an oracle held to the same real-time commitments recovers almost everything the deployed rule loses. We therefore recast the task as sequential selection over the candidates the detector already emits, combining a parameter-free transition model with a set-based scorer that judges each candidate against its competitors in the same instant. This sets a new best on real piano recordings, holds across three independently trained detectors, and shows that what the decision needs is its own temporal history rather than a better image representation.

Index Terms— Score following, sheet music, audio-to-score alignment, real-time, music information retrieval

1. INTRODUCTION

Score following [1] is the online alignment of an acoustic signal to the score being performed. Given the audio observed up to frame t and a representation of the score, the system emits an estimate of the position the performer currently occupies, and it does so under a strict latency budget, in that the estimate for frame t must be produced before frame $t + 1$ arrives. The task supports automatic page turning [2], live computer

accompaniment [3, 4, 5] and score-synchronised visualisation.

Aligning a performance to its score offline is a different problem, usually called audio-to-score alignment or music synchronisation. There the entire recording is available, a global method such as dynamic time warping settles every position at once [6], and even repeats and jumps can be resolved after the fact [7, 8]. Offline alignment underpins cross-modal retrieval and piece identification, where a recording is matched against a collection of scores [9]. Score following, also called music tracking [10], has none of these advantages. It observes only what has been played, commits to a position immediately, and never revises it. Everything below respects this constraint, including the oracles we use to measure what is achievable.

The remaining difficulty is that the two sides are different kinds of object. The performance is audio, the score is an image, and there is no shared space in which to compare them, a mismatch the literature answers in three ways.

1.1. Related work

One family converts the score, using optical music recognition to turn the page into symbols so that a classical follower can track the performance over a discrete set of score states. This line is well developed, spanning online dynamic time warping [6], duration-focused probabilistic models [11], hidden semi-Markov models that absorb performer errors and arbitrary repeats [12], particle filters for rests and tempo change [13], switching state-space models with hidden tempo [14], and multi-agent tracking for robustness to arbitrary deviation [10]. Its weakness is the conversion, since optical music recognition remains error-prone on real scores [15] and the follower inherits every mistake.

A second family converts the audio instead, transcribing the performance into notes and comparing symbol to symbol. Recent work pairs real-time piano transcription with symbol-level tracking and reports strong accuracy [16], though the follower is then bounded by the transcriber, which restricts it to instruments transcription handles well.

The third family converts neither side and learns the correspondence between audio and pixels directly. Dorfer et al. [17] first classified position into staff-level buckets on image crops; reinforcement learning then framed tracking as an agent navigating a score strip [18, 19]; CUNet [20] moved to

full pages, emitting an audio-conditioned segmentation whose centre of mass gives the position; and CYOLO [21] recast it as audio-conditioned object detection, with CYOLO-SB [22] adding heads for the enclosing system and bar. The concurrent CODA [23] cascades system, bar and note selection over the layout regions of an annotated score, with beam search and learned transition penalties.

Training any of these needs performances aligned to note-head coordinates on the page, an annotation costly enough that nearly all of it is synthesized. A score is rendered to MIDI and the MIDI to audio, and the alignment follows from the rendering at no additional cost. The result is systems that work on synthesized audio and lose much of their accuracy on a real instrument in a real room. CUNet [20] falls from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [22] from 89.3% to 79.9%, and CODA [23] from 95.5% to 88.3%. Closing this gap is the field’s central open problem, and every remedy proposed for it has been a perception remedy, whether reverberant augmentation [21], extra proprietary recordings [22], or a new architecture [23].

1.2. Diagnosis

We tested that premise before accepting it, starting from the observation that a conditional detector does not produce a position directly but scores thousands of candidate noteheads and reports the most confident one. Keeping the 256 most confident, we find that on real room-microphone recordings a candidate within the 0.5 s tolerance is present at every one of 4,149 scored onsets. An oracle free to pick any of them is therefore perfect, but could not be run live. The one that matters must commit at each instant and never move backwards, exactly as a deployed follower must, and it still reaches 99.6%. The deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the real-time constraint and are recoverable from what the model already saw, which indicates that reverberation degrades the detector’s confidence ordering rather than the underlying visual evidence.

We therefore recast sheet-image score following as a sequential decision over the positions the page affords, rather than the regression of a coordinate. CANDOR makes that decision with two parts, a transition model scoring a displacement against how much time has passed, and a set-based scorer judging each candidate against its competitors in the same instant, trained listwise on synthesized audio alone. The decision is causal and greedy, with no lookahead and no beam, and costs 1.25 ms per instant on a single CPU thread against the 50 ms available at 20 frames per second.

Our contribution is to relocate the synthetic-to-real gap in this task and to act on it. Candidate-set oracles place the gap in the decision rather than in perception, and CANDOR, a decision model built on that finding, sets a new best on real recordings by a margin that holds across three independently trained perception models.

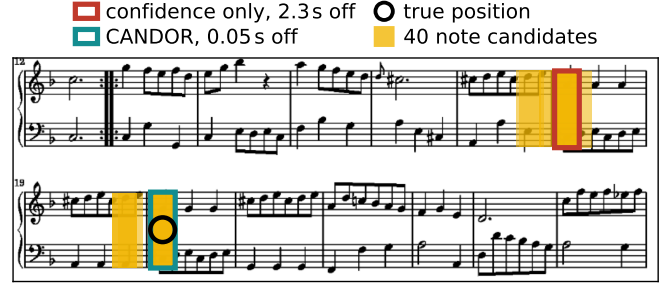


Fig. 1. One instant of a real room-microphone recording. In yellow, 40 of the candidate positions the detector scores. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring). The correct position was available and was not chosen, which is the characteristic failure on real audio.

2. METHOD

2.1. Backbone

Every sheet-image follower of the last five years shares one perception module, a convolutional network over the score page whose feature maps are modulated by the audio. The mechanism is feature-wise linear modulation [24], which injects an external vector into a network by scaling and shifting its activations,

$$f_{\text{FiLM}}(x) = s(z) \odot x + t(z), \quad (1)$$

where s and t are learned linear maps of a conditioning vector z and $\odot$ is elementwise. The page alone cannot say where the performer is, so z carries the recent audio and the visual features become a function of what has just been heard.

CYOLO [21] encodes the last two seconds of audio into z with a spectrogram network and places FiLM layers through the downscale and upscale blocks of a YOLO detector, which then predicts a bounding box for the notehead currently sounding. CYOLO-SB [22] keeps that arrangement and adds heads for the enclosing system and bar, so one 52×52 feature map yields all three. CODA [23] conditions its own backbone the same way, at the deeper stages, before cascading selection over layout regions supplied with the score.

We take this module as given rather than as a design choice, and use the released CYOLO detectors to propose noteheads. No detector weight is updated anywhere in this work, and three separately trained instances are used, so the conclusions concern the family and not one checkpoint.

2.2. CANDOR

CANDOR, for candidate ordering, sits after the backbone and replaces only its final step. Where the detector reports the single most confident box, CANDOR treats the boxes as a candidate set and orders them by how well each continues the

performance so far, carrying its own state from one instant to the next.

2.2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$, and at each one the backbone reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant and report

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i, \quad (2)$$

which is the degenerate selector. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$, and nothing else.

2.2.2. Transition model

Selections must form a plausible trajectory, and between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (3)$$

with $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

2.2.3. Set-based scorer

The transition model treats candidates independently given the history, so what it cannot express is comparative, whether this box, among those on offer in this instant, looks like a real notehead rather than a spurious response. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$ holding its confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the backbone emits, and a neighbourhood group describing how crowded and how typical the box is among its neighbours. We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [25], as in learned re-scoring of detections [26], embeds the set and pools a context shared by all candidates,

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (4)$$

$$s_i = r([e_i; c]), \quad (5)$$

with g and r two-layer perceptrons (64, 32 and 64, 1) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors.

2.2.4. Training and inference

The decision is supervised listwise, so with ε_i the error in frames of candidate i the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (6)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [27]. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (7)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 7 at $\beta = 0.5$ and eliminates all four constants.

3. EXPERIMENTAL SETUP

3.1. Dataset

The decision model is trained on 353 pieces of MSMD [9], synthesized and convolved with measured room impulse responses [28], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). The evaluation set is the real-performance benchmark of [19]: 16 pieces, 25 pages and 4,149 onsets on a Yamaha AvantGrand N2 captured with a room microphone. A held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR.

We report the ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [29].

The four constants (μ, σ, J, β) are fitted leave-one-piece-out, so for each evaluation piece they are chosen on the other 15, pieces weighted equally, so none influences its own decoding, and the residual variant needs no fit at all. Seeds move results by about a point, so rows are seed means with deviations.

4. RESULTS

Table 1 sets CANDOR against the published literature on the same recordings. It tracks 94.4% of onsets within 0.5 s, where the previous best was 88.3%, and leads at every threshold. The margin is not an artifact of one perception model. Retrained on whichever candidates each model emits, the decision gains 20.4, 10.7 and 7.9 points on three of them, and on the weakest, whose training data is entirely public, it passes the point at which the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On the 94 synthetic test pieces CANDOR reaches .955 where

Table 1. Real piano recordings, 16 pieces and 4,149 onsets, Yamaha AvantGrand N2 captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] perception model trained with additional proprietary recordings, public checkpoint. Layout means system and bar annotations required at test time. Δ is the gain at 0.5 s over the perception model the row is built on. Published figures as reported in [23].

Method	Onset error threshold (s)					Layout	Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0		
MM-Loc [17]	.364	.406	.585	.611	.735	no	
RL agent [19]	.185	.200	.476	.603	.901	no	
CUNet [20]	.113	.125	.224	.266	.443	no	
CYOLO [21]	.563	.581	.712	.749	.919	no	
CYOLO-SB [22]	.610	.630	.799	.832	.960	no	
CYOLO-SB+A [†] [22]	.682	.706	.865	.891	.981	no	
CODA [23]	.721	.743	.883	.914	.987	yes	
CANDOR on CYOLO	.726	.747	.916	.950	.991	no	+20.4
CANDOR on CYOLO-SB	.690	.714	.906	.942	.982	no	+10.7
CANDOR on CYOLO-SB+A [†]	.760	.785	.944	.969	.996	no	+7.9

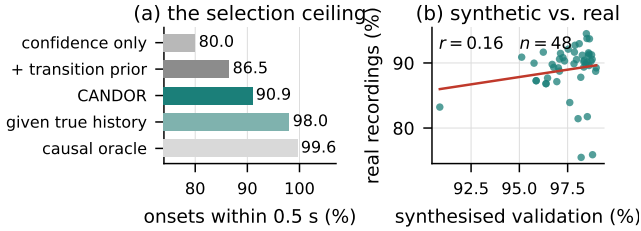


Fig. 2. (a) Every step from the deployed rule to the ceiling on the real recordings; CANDOR to the row given the true history is drift, above it is ranking. (b) Each point is one of 48 trained models, accuracy on synthesised validation audio against accuracy on the real recordings.

argmax reaches .893, and on 80 held-out pieces under a real impulse response and noise it beats the transition model at every level from 12 to 0.5 dB.

The tightest thresholds qualify this, since on CYOLO-SB candidates CANDOR reaches .690 and .714 at 0.05 and 0.1 s, below CODA’s .721 and .743, because a decision is only as precise as the boxes it chooses between, so the mechanism recovers which notehead is active rather than the position within it.

Table 2 separates the parts: the transition model alone, with no learned parameters, accounts for roughly half the gain, and the residual variant, which fits no constant to any evaluation data, still reaches 88.7%.

4.1. Ablation study

Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [30] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so those features are learnable

Table 2. Components of CANDOR, cross-validated per piece on the real recordings at 0.5 s. argmax is Eq. 2 over the same candidates, +transition adds Eq. 3, full adds the scorer.

	n	argmax	+transition	full
<i>Perception model</i>				
CYOLO [21]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [22]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates</i>				
residual, no fitted constants	3	80.0	86.5	88.7 $\pm$ 0.9
without temporal history	3	80.0	86.5	83.7 $\pm$ 0.4
<i>Image features f_i read by the scorer</i>				
FiLM-conditioned backbone	6	80.0	86.5	90.6 $\pm$ 0.9
DINOv2 [30]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9

and useful there and worth nothing on real audio. We rebuilt the perception model with a different audio encoder twice: a frozen MERT-95M tower [31] with only a projection trained, which reaches 46.2 against 90.6 and overfits 12.6-fold on the 353 training pieces, and the two-layer causal Mamba of [23] trained from scratch as they train it, which reaches .. The two are not matched on pretraining and fail differently, on generalisation and on capacity, arriving at the same place. Replacing the visual stem with frozen DINOv2-base patches on the same 52×52 grid, against a control retrained identically, gives ..

Removing the features computed against the previous decision costs 7.2 points, the largest single ablation here. Teacher forcing attributes two thirds of the remaining error to drift, since given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6% (Fig. 2a), which locates the remaining difficulty in the sequence of decisions rather than in the image.

Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real

Table 3. Recovery from 41 discontinuities spliced into the real recordings, on CYOLO-SB+A candidates. rec@1 is the fraction first tracked correctly within 1 s, post the onsets tracked within 1 s over the following 5 s.

Decision rule	break present		no break	
	rec@1	post	rec@1	post
argmax (2)	0.27	0.27	0.15	0.26
CANDOR	0.00	0.08	0.00	0.09
+ silence trigger	0.15	0.17	0.00	0.09
+ confidence trigger	0.10	0.26	0.02	0.23

recordings (Fig. 2b); the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set correlates more strongly ($r = 0.80$) yet still selects poorly. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

4.2. Score discontinuities

Eq. 3 assumes the performance moves forward, which is why the decision is tractable and also why a repeat or a jump breaks it. Following the protocol of [23] we splice three discontinuities into each performance at random destinations, re-running perception on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where state resets anyway, leaving 41.

Table 3 reports a systematic failure, in which after a jump CANDOR performs considerably worse than the rule it replaces, 0.08 against 0.27 of onsets tracked over the next five seconds, because the transition model holds the trajectory at a stale position. Suppressing it on detected silence helps where a break exists and not otherwise, as its design implies; triggering on the scorer’s own confidence needs no break and recovers most of the loss, but never beats argmax; the same ordering holds over 220 jumps on the synthetic benchmark. A transition model strong enough to be worth ten points on continuous playing is strong enough to be wrong at a discontinuity, and reconciling the two is open.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a failure of perception and answered with better audio, more data and new architectures. The measurement reported here places it elsewhere. On real room-microphone recordings the detector already proposes a correct position at every scored onset, and an oracle held to the same real-time commitments recovers almost all of what the deployed rule loses, so the information needed to follow the performance survives the room and what fails is the rule that reads a position out of it.

Acting on that, CANDOR orders the candidates the detector already emits and takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the best published system, at 1.25 ms per instant on a single CPU thread. The margin holds across three independently trained perception models, which suggests the finding concerns the family of audio-conditioned detectors rather than one checkpoint, and it is obtained without retraining any of them and without layout annotation at test time.

What the decision needs is its own temporal history rather than a better image representation, since three image representations including none at all give the same real-audio accuracy while removing the history costs more than any other ablation we ran. Two results temper this. Accuracy on synthesized audio barely predicts accuracy on real audio across 48 trained models, which makes model selection on a synthetic proxy unreliable for the whole field and not only for us. And the forward-motion assumption that makes the decision tractable is what fails at a repeat, where CANDOR is worse than the rule it replaces and no recovery mechanism we tried improves on that rule. Reconciling continuous tracking with discontinuity remains open, and pitch agreement between a candidate and the audio is the natural next source of evidence independent of the history.

Limitations. The correct page is given, all performances are piano, and 16 evaluation pieces cannot resolve differences of a point.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

7. ACKNOWLEDGMENTS

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